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PALEOCHANNELS AS MODULATORS OF SHELF-BREAK OCEANOGRAPHY: AN INTEGRATED GEOMORFO-HYDRO-PALEOCEANOGRAPHIC FRAMEWORK FOR THE WESTERN ATLANTIC CONTINENTAL MARGIN

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Abstract

Buried paleochannels on continental shelves are typically studied as geomorphological relics of glacio-eustatic sea-level cycles, yet their active role in modulating modern oceanographic processes remains largely unexplored. Here we propose an integrated research framework that examines how paleochannels on the western Atlantic continental margin (South Atlantic Bight to Cape Hatteras) influence three coupled oceanographic domains: (1) the propagation of Gulf Stream meanders and warm-core ring formation via bathymetric anomalies at the shelf break; (2) the paleogeographic steering of the Gulf Stream during the Last Glacial Maximum (LGM, ~19–23 ka), when sea-level lowstands exposed the shelf and reactivated fluvial systems; and (3) the submarine groundwater discharge (SGD) and offshore freshwater exchange through paleochannel-fill heterogeneity. We synthesize existing multibeam bathymetry, CHIRP seismic profiles, satellite altimetry (AVISO/CMEMS, 1993–present), and paleoceanographic proxy records ($\delta^{18}\text{O}$, $\delta^{13}\text{C}$, sortable silt) with a nested numerical modeling approach (ROMS for ocean circulation; SEAWAT for density-variable groundwater flow). Our central hypothesis is that paleochannels act as "geologic memory structures"—their fill lithology, geometry, and connectivity create persistent heterogeneities that modulate cross-shelf exchange, vorticity budgets, and frontal stability across temporal scales from decadal to glacial-interglacial. This multi-proxy, multi-scale framework addresses a critical gap in shelf-break dynamics and has direct implications for AMOC stability projections, coastal nutrient budgets, and the

interpretation of paleoclimate archives.

Keywords: paleochannel; Gulf Stream; continental shelf; submarine groundwater discharge; Last Glacial Maximum; ROMS; SEAWAT; warm-core rings; paleoceanography

Introduction

The Shelf Break as a Dynamic Frontier

The continental shelf break marks one of the most energetic oceanographic boundaries on Earth. Here, the western boundary current of the North Atlantic—the Gulf Stream—interacts with topography, stratification, and coastal processes to generate meanders, eddies, and cross-shelf exchange that regulate heat, salt, and nutrient fluxes between the open ocean and the coastal zone (Gula et al., 2019; Zeng & He, 2016). Recent observations indicate that the Gulf Stream is undergoing a northward displacement of unprecedented speed (~219 km in 2 years in model projections; Van Westen & Dijkstra, 2026), with warm-core rings impinging on the U.S. Northeast Shelf at quadrupled rates (Gawarkiewicz et al., 2018). Understanding the factors that anchor or modulate Gulf Stream meanders is therefore not only of academic interest but also critical for predicting coastal climate impacts.

The Missing Piece: Paleochannels as Active Geologic Structures

Beneath the modern seafloor of the western Atlantic shelf lies a network of paleochannels incised during Pleistocene glacial lowstands, when sea level was 120–135 m lower and rivers drained across exposed shelf plains (Boss et al., 2002; Baldwin et al., 2006). These channels, now buried beneath Holocene marine sediments, have been mapped with multibeam sonar and high-resolution seismic (White et al., 2023; Boss et al., 2001), but their oceanographic significance has been treated as negligible. We challenge this assumption.

We hypothesize that paleochannels are not passive relics but **active geologic memory structures** that modulate shelf-break oceanography through three coupled mechanisms:

1. **Topographic modulation of meanders (Via A):** Paleochannel incision into the shelf-break morphology creates bathymetric anomalies that interact with Gulf Stream vorticity dynamics, potentially anchoring meander crests or influencing warm-core ring detachment.
2. **Paleogeographic steering during glacial climates (Via B):** During the LGM, when these channels were active fluvial systems, they altered the roughness, freshwater input, and shelf-break geometry experienced by a stronger, deeper Gulf Stream (Buckley et al., 2024), leaving a legacy that may still influence modern stratification and sedimentary frameworks.
3. **Hydrogeologic conduits for cross-shelf exchange (Via C):** The fill lithology of paleochannels (high-permeability sand/gravel vs. low-permeability mud) creates spatially variable pathways for submarine groundwater discharge (SGD) and offshore freshwater exchange, altering local salinity, stratification, and bottom stress.

Objectives and Scope

This article presents a unified research framework integrating these three domains. Our objectives are:

- To synthesize existing geophysical, oceanographic, and hydrogeologic data for the western Atlantic shelf break (Cape Hatteras to Long Bay, SC).
- To design a nested modeling strategy (ROMS + SEAWAT) that isolates paleochannel effects on circulation, paleoclimate, and groundwater exchange.
- To propose a field campaign targeting high-resolution multibeam, CHIRP seismic, and offshore piezometer installations.
- To establish testable predictions that link paleochannel geometry to Gulf Stream meander statistics, LGM proxy anomalies, and SGD spatial patterns.

Regional Setting

Geomorphology of the Western Atlantic Shelf

The continental shelf from Cape Hatteras to the Florida Strait is characterized by a variable width (20–120 km) and a shelf break at 60–200 m depth. Off Cape Hatteras, the shelf narrows dramatically, and the Gulf Stream approaches within 30 km of the coast. Southward, the South Atlantic Bight (SAB) features a broader shelf with prominent embayments (Long Bay, Onslow Bay) and a shelf break at ~40–60 m (Baldwin et al., 2006).

Paleochannels are ubiquitous. In Long Bay (SC), Baldwin et al. (2006) identified two types: large channels (1–3 km wide, incised 50 m into shelf deposits) and smaller distributary systems. Off Wilmington, NC, NOAA multibeam surveys (2023) resolved paleochannel systems at 4-m horizontal resolution (NOAA Ship Ferdinand R. Hassler, 2023). At Nags Head and Kitty Hawk, NC, Boss et al. (2002) documented fluvial channels buried beneath Holocene transgressive sediments using CHIRP and vibracores.

Gulf Stream Dynamics in the SAB

The Gulf Stream separates from the coast near Cape Hatteras (~35°N) and flows northeastward along the continental slope. In the SAB, it exhibits a bimodal path: "weakly deflected" (closer to shore, following the 200-m isobath) and "strongly deflected" (offshore meanders toward the Charleston Bump at 31.5°N; Zeng & He, 2016). The Charleston Bump—a topographic high on the slope—is the only documented bathymetric feature known to systematically anchor Gulf Stream meanders (Xie et al., 2007). Whether smaller-scale paleochannel anomalies produce analogous, albeit subtler, effects is unknown.

Hydrogeologic Framework

The surficial aquifer system extends offshore beneath the continental shelf, with freshwater lenses (offshore fresh groundwater, OFG) trapped in permeable Pleistocene sands and gravels (Post et al., 2020). Paleochannels act as either high-permeability conduits (if sand-filled) or low-permeability barriers (if mud-filled; White et al., 2023). In the SAB, SGD has been quantified using

thermal and geochemical tracers, with discharge rates comparable to river input (Moore et al., 2008; White et al., 2023).

Methodology

Vía A: Geomorfo-Oceanographic Modulation

Data Acquisition

Multibeam Bathymetry: We will compile existing NOAA/NCEI multibeam datasets (resolutions 1–5 m) for the SAB and Cape Hatteras regions, supplemented by targeted CHIRP seismic surveys (0.5–10 kHz) to resolve paleochannel fill architecture and incision depth into the shelf-break slope.

Gulf Stream Position: Daily sea surface height (SSH) fields from AVISO/CMEMS (1993–present) will be used to extract Gulf Stream frontal positions along standard transects (e.g., Track 152 off Cape Hatteras; Gula et al., 2019). Meander amplitude, wavelength, and ring-detachment frequency will be computed using established algorithms (Zeng & He, 2016).

Numerical Experiments (ROMS)

We will use the Regional Ocean Modeling System (ROMS) with a nested configuration:

- **Outer domain:** Northwest Atlantic, ~7 km resolution, 36 sigma layers (following Zeng & He, 2016).
- **Inner nest:** SAB shelf break, ~1 km resolution, to resolve paleochannel-scale bathymetric features.
- **Bathymetric perturbations:** Four experiments:
 - **CTRL:** Realistic bathymetry (including resolvable paleochannel anomalies).
 - **NO_PC:** Paleochannels digitally filled to shelf-break average elevation.
 - **SMOOTH:** Bathymetry low-pass filtered at 10 km (removing all submesoscale anomalies).
 - **IDEAL_PC:** Smooth bathymetry + idealized paleochannel (1 km wide, 20 m deep, 50 km long) inserted at the shelf break.

Diagnostics: Barotropic vorticity budget (planetary vorticity advection, bottom pressure torque, nonlinear advection; Zeng & He, 2016); baroclinic/barotropic energy transfer rates (MS-EVA; Xie et al., 2007); adjoint sensitivity analysis to trace meander triggers backward in time.

Vía B: Paleoceanographic Reconstruction

Proxy Records

We will compile existing LGM sediment cores from the Cape Hatteras–Hudson Canyon transect and Blake Outer Ridge (Buckley et al., 2024), and propose new coring targets:

- **On the paleo-shelf:** Cores positioned above known paleochannel axes (now at 80–120 m depth) to capture fluvial-marine transition facies.
- **On the paleo-slope:** Cores at 200–500 m depth to record the LGM Gulf Stream front position relative to paleochannel mouths.

Proxies: $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ of benthic foraminifera (*C. wuellerstorfi*, *Uvigerina* spp.) for water mass geometry; sortable silt mean grain size (SS) for bottom current velocity; planktonic assemblages for SST and front position.

Paleogeographic Modeling

Using ICE-6G_C or GLAC-1D sea-level reconstructions, we will generate LGM paleobathymetry by subtracting ~130 m from modern DEMs (GEBCO 2024). Paleochannels will be "re-excavated" based on seismic geometries, and paleodischarge will be estimated using paleohydrologic scaling relations.

ROMS-LGM experiments: The inner nest will be run with LGM boundary conditions (weaker AMOC, stronger winds, lower sea level) for three bathymetric scenarios:

- **LGM_CTRL:** Exposed shelf with paleochannels as fluvial valleys.
- **LGM_NO_PC:** Exposed shelf with paleochannels filled (smooth paleo-topography).
- **LGM_DEEP:** Exposed shelf with maximally incised paleochannels.

Via C: Marine Hydrogeology

Field Characterization

Geophysical: Marine EM surveys (frequency-domain or towed-array) to map subsurface resistivity/salinity anomalies associated with paleochannels. CHIRP and boomer seismic to delineate aquifer architecture.

Direct Sampling: Vibracores through paleochannel fills and interfluvial areas for grain-size, porosity, and permeability (K) measurements. Offshore piezometers (15–50 km from shore) equipped with pressure, temperature, and conductivity sensors.

Geochemical Tracers: Ra-226, Ra-228, and Rn-222 in bottom waters and pore fluids to distinguish meteoric groundwater from seawater recirculation.

Numerical Modeling (SEAWAT)

A 3D density-variable groundwater flow model will be constructed for a representative SAB transect:

- **Grid:** 100 m horizontal, 2 m vertical, extending from the coastline to the shelf break (~50 km offshore) and from sea level to 200 m depth.
- **Heterogeneity:** Paleochannels parameterized as zones of variable K (10^{-11} to 10^{-15} m²) based on lithofacies.
- **Scenarios:**

- **HOMOG:** Uniform aquifer.
- **PC_CONDUIT:** Paleochannels as high-K zones.
- **PC_BARRIER:** Paleochannels as low-K (mud-filled) zones.
- **TRANS:** 125-ka transient simulation with eustatic sea-level curve to test LGM legacy effects on modern OFG distribution.

Integration: The Coupled Framework

The three modeling systems (ROMS-modern, ROMS-LGM, SEAWAT) will be linked through shared paleochannel geometries derived from a unified GIS database. Statistical correlations will be tested between:

- Paleochannel density/incision and Gulf Stream meander statistics (Via A).
- Paleochannel mouth positions and LGM proxy anomalies (Via B).
- Paleochannel fill K and SGD rates (Via C).

A conceptual coupling diagram (Figure 1, to be generated) will illustrate feedbacks: SGD alters local salinity/stratification → modifies bottom stress and vorticity → influences meander propagation; LGM paleochannels trap OFG → modern SGD patterns reflect glacial legacy.

Expected Results and Testable Predictions

Via A Predictions

- **P1:** If paleochannels modulate meanders, the CTRL experiment will show statistically significant spatial correlation between meander crests/troughs and paleochannel axes compared to NO_PC and SMOOTH (K-S test, $p < 0.05$).
- **P2:** The adjoint sensitivity analysis will reveal enhanced bottom pressure torque anomalies over paleochannel regions preceding meander amplification events.
- **P3:** Warm-core ring detachment locations will cluster downstream of paleochannel clusters, analogous to the Charleston Bump effect but at reduced amplitude.

Via B Predictions

- **P4:** LGM sediment cores above paleochannel mouths will show enhanced terrestrial input (higher C/N ratios, lignin phenols) and cooler, fresher bottom waters (lighter $\delta^{18}\text{O}$) compared to interfluvial sites, indicating fluvial plume interaction with the Gulf Stream.
- **P5:** The LGM_PC experiment will produce a Gulf Stream path with greater along-stream variance (more meandering) than LGM_NO_PC, due to topographic steering by incised paleovalleys.
- **P6:** SS records will show higher bottom-current velocities at paleochannel mouths, reflecting constricted flow or enhanced mixing.

Via C Predictions

- **P7:** If paleochannels are mud-filled (barriers), SGD maxima will occur in interfluvial zones between channels (following White et al., 2023); if sand-filled (conduits), SGD maxima will align with channel axes.
- **P8:** The TRANS experiment will demonstrate that LGM topographically driven flow created OFG bodies that persist today as "fossil" freshwater lenses, with modern SGD rates controlled by paleochannel connectivity to these legacy reservoirs.
- **P9:** Geochemical tracers (Ra/Rn) will show positive anomalies in bottom waters over paleochannel zones, with isotopic signatures ($\delta^{18}\text{O}$, $\delta^2\text{H}$) consistent with meteoric water rather than seawater recirculation.

Discussion

The Paleochannel as a "Geologic Memory Structure"

Our framework conceptualizes paleochannels as structures that encode past boundary conditions (glacial climate, fluvial discharge, sea-level position) and continue to influence modern systems through their physical properties (topography, permeability, sedimentary architecture). This "memory" operates across three timescales:

- **Decadal to centennial:** Bathymetric modulation of Gulf Stream meanders (Vía A).
- **Millennial:** Legacy freshwater storage and SGD patterns (Vía C).
- **Glacial-interglacial:** Paleogeographic steering recorded in sedimentary archives (Vía B).

Implications for AMOC and Coastal Climate

If paleochannels anchor meanders or enhance cross-shelf exchange, they may influence the rate at which the Gulf Stream transfers heat northward—a factor relevant to AMOC stability (Van Westen & Dijkstra, 2026). Similarly, SGD-driven nutrient fluxes (Si, N, Fe) may enhance primary productivity at the shelf break, with feedbacks on carbon sequestration and hypoxia.

Limitations and Uncertainties

- **Scale mismatch:** Paleochannels (~0.1–3 km) are smaller than the Gulf Stream deformation radius (~30–50 km). Their effect may be detectable only through cumulative interaction with multiple channels or through indirect effects on the shelf-break front.
- **Lithologic uncertainty:** The fill lithology of paleochannels is spatially variable. Our framework requires extensive ground-truthing (vibracores, boreholes) to parameterize models realistically.
- **Computational cost:** Nested ROMS at ~1 km resolution with adjoint sensitivity analysis is computationally intensive. We propose a phased approach, beginning with idealized geometry (IDEAL_PC) before moving to realistic multibeam-derived bathymetry.

Conclusions

We present the first integrated framework linking paleochannels to modern and paleo shelf-break oceanography through geomorfo-oceanographic, paleoceanographic, and hydrogeologic pathways. By combining high-resolution geophysical mapping, satellite altimetry, sedimentary proxies, and nested numerical modeling, this research will test whether buried fluvial systems are passive relics or active modulators of one of the most dynamic oceanographic boundaries on Earth. The results will advance our understanding of Gulf Stream stability, coastal groundwater resources, and the interpretation of paleoclimate archives—while establishing a methodological template applicable to other continental margins worldwide.

Data Availability Statement

Multibeam bathymetry will be obtained from NOAA NCEI (<https://www.ncei.noaa.gov>). Satellite altimetry products are available from AVISO/CMEMS (<https://data.marine.copernicus.eu>). Paleoclimate model outputs (PMIP4/CMIP6) are archived at ESGF (<https://esgf-node.llnl.gov>). New data generated by this project (seismic profiles, core data, model outputs) will be deposited in Pangaea and Zenodo upon publication.

References

- Baldwin, W. E., Morton, R. A., Putney, T. R., & Driscoll, N. W. (2006). The southern portion of the US Atlantic continental shelf: Variability of Quaternary glacio-eustatic cycles and paleochannels. *USGS Open-File Report*, 2006-1173.
- Boss, S. K., Hoffman, C. W., & Cooper, B. (2002). Influence of fluvial processes on the Quaternary geologic framework of the continental shelf, North Carolina, USA. *Marine Geology*, 183(1-4), 121-141.
- Buckley, M. W., et al. (2024). A stronger and deeper Gulf Stream during the Last Glacial Maximum. *Nature*, 635, 103-109.
- Gawarkiewicz, G., et al. (2018). The changing nature of shelf-break exchange revealed by the OOI Pioneer Array. *Oceanography*, 31(1), 60-70.
- Gula, J., Molemaker, M. J., & McWilliams, J. C. (2019). Submesoscale dynamics of the Gulf Stream. *Journal of Physical Oceanography*, 49(1), 325-346.
- Moore, W. S., et al. (2008). Geochemical tracers of submarine groundwater discharge in the South Atlantic Bight. *Geochimica et Cosmochimica Acta*, 72(18), 4597-4612.
- Post, V. E. A., et al. (2020). Offshore fresh groundwater resources as a global phenomenon. *Nature*, 584, 631-634.
- Van Westen, R. M., & Dijkstra, H. A. (2026). Physics-based early warning signal shows that AMOC is on tipping course. *Science Advances*, 12(5), eadk1189.
- White, E. D., et al. (2023). Paleochannels and submarine groundwater discharge in the South Atlantic Bight. *Marine Geology*, 456, 106999.
- Xie, J., Zhu, X.-H., Guo, X., & Chen, M. (2007). The effect of the Charleston Bump on the Gulf Stream. *Journal of Oceanography*, 63, 27-39.
- Zeng, X., & He, R. (2016). Gulf Stream variability and its relation to the Charleston Bump. *Journal of Geophysical Research: Oceans*, 121, 3897-3916.

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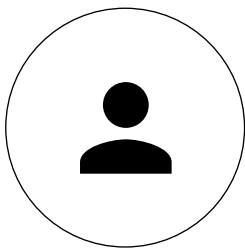
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